

Preliminary Examinations 2026

XII (SCIENCE P.M & P.E) CHEMISTRY

SECTION 'A'

Multiple Choice Questions (MCQs)

17 Marks

Time: 20 minutes

Note: Attempt all parts from this section.

Q.1: Choose the correct answer for each from the given options.

- (i) The space between 0 - 11 km above the surface of the Earth is called _____ :
 (a) Troposphere (b) Lithosphere (c) Biosphere (d) Hydrosphere
- (ii) Scurvy is caused by the deficiency of _____ vitamin:
 (a) A (b) B (c) C (d) D
- (iii) "A" belongs to IIIA group while "B" belongs to IVA, The formula of the compound formed from these elements would be _____ :
 (a) A_2B_3 (b) A_3B_4 (c) A_4B_3 (d) A_3B_2
- (vi) Most of the properties of Li resemble with the properties of _____ :
 (a) Mg (b) Ca (c) Cu (d) Na
- (v) Large scale manufacture of sulphuric acid is carried out by _____ method
 (a) Contact (b) Down's (c) Ostwald (d) Castner Killner
- (vi) Benzene gives electrophilic substitution reactions due to the presence of _____
 (a) Electrons (b) planar structure (c) unhybridized orbitals (d) Aliphatic nature
- (vii) Ethylenediamine is _____ legend:
 (a) Bidentate (b) Neutral (c) Organic (d) All of these
- (viii) The polypeptides of different amino acids _____ :
 (a) Proteins (b) Carbohydrates (c) Vitamins (d) Fats and oils
- (ix) I.U.P.A.C. name of $[Ni(CO)_4]$ is _____ .
 (a) Nickel carbonyl (b) Tetracarbonyl Nickel
 (c) Tetracarbonyl Nickel(0) (d) Nickel carbonmonoxide
- (x) The compounds of _____ family have fruity smell:
 (a) Alcohol (b) Aldehyde (c) Ether (d) Ester
- (xi) When Formaldehyde is reacts with Fehling's solution, Orange precipitates of _____ are formed:
 (a) CuO_2 (b) CuO (c) Cu_2O (d) Cu_2O_2
- (xii) _____ structure of benzene removes the objections raised on Kekule's structure of benzene.
 (a) Kekule (b) Molecular orbital (c) Cyclic (d) Resonating
- (xiii) Which of the following is meta director:
 (a) $-NH_2$ (b) $-CH_3$ (c) $-COOH$ (d) $-Cl$
- (xiv) _____ of the following hydrocarbons shows acidic properties:
 (a) Methane (b) Ethene (c) Acetylene (d) Ethylene
- (xv) Formaldehyde reacts with Grignard's reagent to form:
 (a) 1^{st} alcohol (b) 2^{nd} alcohol (c) 3^{rd} alcohol (d) Carboxylic acid
- (xvi) _____ is a non-sugar carbohydrate:
 (a) Fructose (b) Glucose (c) Cellulose (d) Sucrose
- (xvii) The homogeneous mixture of some elements in which at least one is metal is called _____ :
 (a) Solution (b) Compound (c) Alloy (d) stainless steel

SECTION 'B'

(Short Answer-Questions)

36 Marks

Note: Answer any Nine part questions. Select Four part questions from Inorganic - General Chemistry and Five part questions from Organic Chemistry. All part questions carry equal marks.

INORGANIC CHEMISTRY

Q.2: Write IUPAC names of the following complexes:

- (a) $K_4[Fe(CN)_6]$ (b) $[Cu(H_2O)_4]Cl_2$ (c) $[Zn(OH)_4]^{2-}$ (d) $[Fe(CN)_4]^{2-}$

(ii) Describe the trends of metallic characters and ionization potential in the periodic table.

(iii) Describe how bleaching powder is manufactured and give its reactions with HCl and H_2O .

(iv) Give reasons of the following: (Any four).

(a) Atomic radii of alkali metals are larger than alkaline earth metals.

(b) Magnesium is harder than sodium. (c) The B.P and viscosity of H_2SO_4 are very high.

(d) Ligands are also called Lewis bases. (e) Alkali metals are powerful reducing agents.

(v) What information about the structure of a molecule we can get from NMR spectroscopy? Give one example.

- (vi) What is meant by green house effect? What are green house gases? How they cause global warming?
 (vii) Write a note on any two: Chromium, potassium dichromate, manganese, potassium permanganate.

ORGANIC CHEMISTRY

(viii) Discuss the Molecular orbital structure of Ethane (OR) Ethene.

(OR) What happens when Ethylene is reacted with the following:

- (a) HBr (b) HOCl (c) Chlorine Water (d) KMnO_4 (e) H_2

(ix) What is Isomerism? And write any five isomers of decanoic acid with their I.U.P.A.C names.

(OR) What happens when Acetylene reacts with AgNO_3 , Na , Cu_2Cl_2 , KMnO_4 , H_2O

(x) What are Aldehyde and Ketone? Give any two reactions with equation to distinguish them. (OR) Write a

Note on natural sources of organic compounds (OR) Distinguish the following by simple chemical test.

- (a) Alkanes and Alkenes (b) Alkanes and Alkynes
 (c) Alkenes and Alkynes (d) Aldehydes and Ketones
 (e) Fats and Oils (f) Alkanes and Alkyl halides

(xi) Benzene behaves like saturated as well as unsaturated compound. Give two examples.

(OR) Describe the Molecular orbital structure of benzene.

(xii) How will you prepare the following? Write only chemical equation

- (a) Formic acid from methanol (b) Ethane from methane
 (c) Ethyl acetate from acetic acid (d) Toluene from benzene

(xiii) Write down the balanced chemical equation for the following.

- (a) Reaction of ethyl iodide with sodium metal.
 (b) Reaction of Propene with HBr .
 (c) Reaction of ethene with alkaline KMnO_4 solution.
 (d) Reaction of ethyne with silver nitrate in the presence NH_3 .

(xiv) Write down the I.U.P.A.C names of the following.

- (a) $(\text{CH}_3)_3\text{C} - (\text{CH}_2)_2 - \text{CHO}$ (b) $\text{CH}_3 - (\text{CH}_2)_3 - \text{CO} - \text{CH}_3$
 (c) $\text{HOOC} - (\text{CH}_2)_4 - \text{COOH}$ (d) $\text{CH}(\text{CH}_3)_2 - \text{CO} - (\text{CH}_2)_2 - \text{COOH}$

(xv) What are alcohols? Describe the classification of monohydric alcohols.

(OR) Write the reactions of Ethyl alcohol with the following (Any four):

- (a) Na (b) PCl_3 (c) $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$ (d) H_2SO_4 (180 °C)
 (e) PCl_5 (f) Ni / Cu (g) CH_3COOH (h) SOCl_2

SECTION 'C'

(Detailed Answer Questions)

32 Marks

Note: Answer any Four questions — Two question from Inorganic – General Chemistry and Two from Organic Chemistry. All questions carry equal marks.

INORGANIC CHEMISTRY

Q.3: Describe the metallurgy of copper from its sulphide ore and the refining of blister copper. (OR)

Q.4: Give the manufacture of caustic soda by Castner – Killner method with advantages and disadvantages

Describe the manufacture of sulphuric acid by contact process along with its properties as an acid, dehydrating agent, oxidizing agent, sulphonating agent. (OR)

Discuss any six properties of transition metals.

Q.5: Define Troposphere and stratosphere. Describe the chemistry involved due to the presence of oxides of carbon and nitrogen in the troposphere. (Write equations where necessary)

ORGANIC CHEMISTRY

Q.5: (a) How is Grignard's reagent prepared? Choose the suitable Grignard's reagent to prepare acetic acid, sec. propyl alcohol and tertiary butyl alcohol with mechanism. (OR) Bring about the following conversions from benzene. (i) 'TN'

- (ii) Benzoic acid (iii) o – nitro benzoic acid (iv) m – dinitrobenzoic acid

(b) Describe E1 and E2 reactions (OR) SN^1 and SN^2 reactions with mechanism in details.

Q.6: (a) Write any two methods of preparation for each of the following:

- (i) Formaldehyde (b) Acetone (c) Acetic acid (d) Phenol

OR What happens when Formaldehyde reacts with the following:

- (i) $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$ (ii) $\text{R} - \text{MgX}$ (iii) Tollen's reagent (iv) Fehling's solution

(b) What are electrophilic substitution reactions? Give the example of Nitration and Sulphonation of benzene with mechanism. OR Write the reaction of Grignard's reagent with the following:

- (i) H_2O (ii) $\text{R} - \text{OH}$ (iii) Formaldehyde (iv) Acetone

OR Write the short note on the following: (Any four)

- (i) Potassium dichromate (ii) Proteins (iii) carbohydrates (iv) Plastics (v) Fats and oils
 (vi) Troposphere (vii) Stratosphere (viii) Alcohols (ix) Esters (x) Saponification